

DataQuest 2026: Strategic Credit Modelling Masterclass

From Raw Data to Interpretable Business Strategy

Dichaba Mofokeng

May 22, 2026

How to Navigate This Program

Project Execution:

- **Setup:** Install dependencies via `pip install -r requirements.txt`.
- **Execution:** Run the command `streamlit run app/main_app.py` in your terminal.
- **Dashboard Logic:**
 - **Tab 1 (EDA):** Explore data quality and identify risk patterns.
 - **Tab 2 (Model):** View the scorecard and performance metrics.
 - **Tab 3 (Business):** Simulate profit and macro-economic stress.
 - **Tab 4 (Fairness):** Perform regulatory bias audits.

The Regulatory Paradox

- **The Conflict:** Banks need high accuracy, but regulators demand total transparency (Explainability).
- **The Constraint:** Black-box models (Neural Networks/Deep Learning) are prohibited for final credit decisions.
- **The Solution:** We use Logistic Regression. To make it perform at the level of complex models, we perform **Weight of Evidence (WoE) Feature Engineering**.

Key Principle: Strategic transformation allows a simple linear model to capture complex, non-linear risk signals without losing transparency.

Weight of Evidence (WoE) Theory

WoE transforms raw features into a risk-based scale by comparing the distribution of Good vs. Bad customers.

The Mathematical Definition

$$WoE_i = \ln \left(\frac{\% \text{ Good Customers}_i}{\% \text{ Bad Customers}_i} \right)$$

- **Positive WoE:** Higher proportion of Goods (The bin is statistically safe).
- **Negative WoE:** Higher proportion of Bads (The bin is statistically risky).
- **Zero WoE:** Neutral risk (Portfolio average).

Quantifying Signal: Information Value (IV)

IV measures the total predictive power of a feature. It is calculated by weighting the difference in distributions by the WoE.

The IV Formula

$$IV = \sum_{i=1}^n (\% \text{ Goods}_i - \% \text{ Bads}_i) \times WoE_i$$

Decision Rules:

- **IV ; 0.02:** Useless (Feature likely noise).
- **0.1 – 0.3:** Medium (Standard model feature).
- **IV > 0.3:** Strong (Primary risk driver).

Advanced Technique: Shadow Modelling

- **The Challenge:** How do we know what patterns to look for?
- **The Technique:** We train a non-linear **Random Forest** purely as a "Shadow Model."
- **The Benefit:** It highlights hidden interactions (e.g., Utilization vs. Income) that we wouldn't see in simple correlation.
- **The Result:** We translate those findings into our WoE bins for the Logistic model.

Scorecard Scaling: Points to Double Odds (PDO)

We translate logistic probabilities into an industry-standard score (e.g., 300 to 900 points).

- **Factor:** Determines how many points the score increases per risk unit.

$$Factor = \frac{PDO}{\ln(2)}$$

- **Offset:** Aligns the score to a specific anchor (e.g., 600 points).

$$Offset = \text{Target Score} - (Factor \times \ln(\text{Target Odds}))$$

Architectural Excellence: Intercept Distribution

A common error is ignoring the model intercept (α). We distribute it across all n features to ensure accuracy.

The Point Calculation Formula

$$Points_{ij} = -(\beta_i \times WoE_{ij} \times Factor) - \left(\frac{\alpha}{n} \times Factor \right) + \frac{Offset}{n}$$

Why this matters: This mathematical step ensures that when a customer's individual attribute points are summed, the total equals the exact probability predicted by the logistic regression.

Business Value: Stress Testing

Models are built on history but deployed in the future.

- **Macro-Economic Stress:** We apply a multiplier to default rates to simulate a recession.
- **P&L Engine:** We translate probabilities into Expected Revenue and Expected Loss:

$$\text{Net Profit} = (\text{Revenue}) - (\text{Defaults} \times \text{Loss Multiplier})$$

Application: This tool allows management to choose the "cutoff" that maximizes profit, not just accuracy.

Stability Monitoring: PSI

The Population Stability Index (PSI) detects "data drift" (when incoming applicants no longer look like the training data).

PSI Formula

$$PSI = \sum (\text{Actual}\% - \text{Expected}\%) \times \ln \left(\frac{\text{Actual}\%}{\text{Expected}\%} \right)$$

Thresholds:

- **PSI ; 0.1:** Stable.
- **PSI ; 0.2:** Major drift; immediate retraining required.